

Is Agriculture Safe From AI?

The already-in-it guide — the full scores by track, the six factors behind them, and what to do from where you already are.

4.0

AI EXPOSURE · FARM MANAGEMENT / MIXED FARMING

where 10 is most at risk

For anyone already in it — in the degree, or job-hunting.



Before you read this

The choice is already made — you're in an agriculture degree or training, or out of one and into the job hunt. So this guide skips the "should you go into agriculture" question and goes straight to the one that's actually live: given where you already are, what do you do about it?

It's written to you directly — whether you are the one in it, or a parent reading on their behalf. Where it says "you," a parent can simply read "them"; the moves are the same either way.

What this is. A facts document, not a pep talk. It sets out what's known about where agriculture sits in an AI economy, where the evidence is contested, and where we might be wrong — and then what to actually do from here.

The good news, up front. Already in it is not too late — not close. Agriculture isn't one destination, it's a wide spread of them, from 3.4 to 7.0, and almost none of your exposure is fixed yet. Where you land inside that range is decided mostly by moves still in front of you — chiefly, whether you steer toward the physical, wage-paying work and away from the analytical layer. The second half of this guide is those moves.

And one thing specific to agriculture. There's one statistic in this guide that reframes the entire conversation, and it has nothing to do with AI: USDA forecasts that the median farm household will lose money farming in 2026, and earn \$92,815 off the farm. That isn't an argument against agriculture. It's an argument that farming and working in agriculture are different careers with different economics — and the economics section below sets out why that distinction should shape what you steer toward from here.

The short answer

Safe for hands-on production work — and agriculture has been living with automation longer than any other career in this index.

Livestock husbandry scores 3.4 out of 10. Field farming and crop production scores 4.0. Precision agriculture technology scores 6.2, and agricultural data and supply planning scores 7.0.

The same inversion appears here as in the life sciences: **the tech-forward version of the career is more exposed than the hands-on one.** So the work now is steering toward the protected end of that spread, not just landing anywhere in it.

And the honest question in agriculture was never automation. It's whether the economics work — which is why *where you aim from here* carries so much weight.

The scores

TRACK	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Livestock husbandry & animal management	2.9	3.2	3.4	+0.5	Low
Agricultural equipment technology & repair	3.2	3.5	3.8	+0.6	Low
Field farming & crop production	3.4	3.7	4.0	+0.6	Low–Moderate
Farm management	3.9	4.3	4.6	+0.7	Moderate
Agronomy & crop advisory	4.5	5.0	5.4	+0.9	Moderate
Precision agriculture technology	5.0	5.7	6.2	+1.2	Moderate–High
Agricultural data & supply planning	5.8	6.5	7.0	+1.2	High

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, a service electrician scores 2.5, bedside nursing 2.8, and a business analyst 7.6.

Read the table this way: the ordering runs from hands and animals at the top to screens and data at the bottom. That is the same pattern as every other profession in this index, and in agriculture it happens to invert the intuition about which jobs are "modern." Your job, from here, is to move up the table toward the field-based end — not to hope the table moves.

Agriculture has been running this experiment for seventy-five years

This is the most useful section in this guide, because agriculture is the longest-running case study of automation in any career we score — and the results are already in.

The record, from USDA's own data:

- Self-employed and family farmworkers fell from **7.60 million in 1950 to 2.06 million by 2000** — a 73% reduction.
- Hired farmworkers fell from **2.33 million to 1.13 million** over the same period — 51%.
- And across the same era, **total farm output nearly tripled between 1948 and 2021 while total labor hours declined.**

Three-quarters of the workforce disappeared while output tripled. That is a more complete displacement than anything currently happening in law, design or software — and it happened over decades, through mechanisation, chemistry, genetics and farm consolidation rather than through anything anyone called AI.

What that history tells us, and it's worth carrying to every other profession:

Automation removed the labor, not the work. Somebody still farms. There are simply far fewer of them, each operating at much larger scale with much more capital.

The survivors got bigger, not safer. USDA's Census of Agriculture found that **105,384 farms with sales over \$1 million — 6% of all US farms — sold more than three-quarters of all agricultural products.** Meanwhile 1.4 million farms with sales under \$50,000 made up 74% of farms and 2% of sales.

And the displacement was invisible from outside. Nobody outside agriculture experienced the loss of five million farm jobs as a crisis, because it happened gradually and food kept arriving.

The general lesson: when a sector automates, the question isn't whether the work survives — it usually does. It's how many people are needed, at what scale, and whether the ones who remain own the capital or operate it. Agriculture answered all three a generation ago, and every profession now facing automation is asking the same questions for the first time.

The inversion: why precision agtech scores worse than farming

The same finding appears in the life sciences guide, and it surprises people equally here.

Precision agriculture is the modern, technical, well-regarded end of the field — GPS guidance, variable-rate application, yield mapping, drone and satellite imagery, sensor networks, decision-support platforms.

And it scores 6.2 against field farming's 4.0. Here's why.

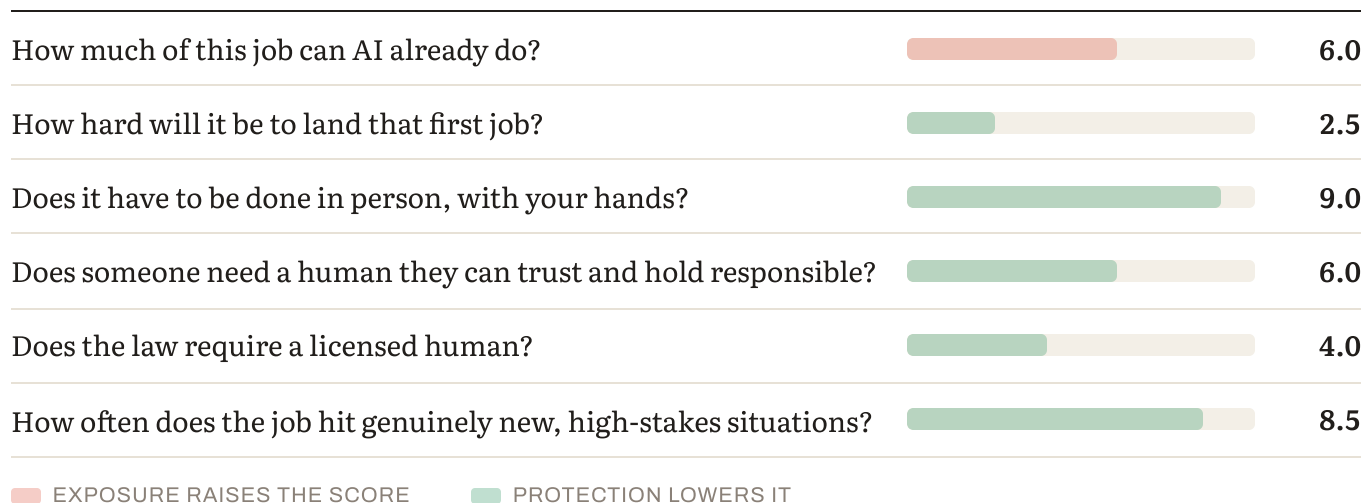
Ask what the work consists of. Precision agriculture takes structured data — soil samples, yield maps, weather, imagery — and produces recommendations. Structured inputs, analytical outputs, no physical presence required, no licensed accountability. That is the exposed profile in every field we score.

Field farming, by contrast, requires being there. Livestock don't cooperate. Weather changes the plan. Equipment breaks in a field at 6am. Soil conditions differ from what the map says. The work is physical, unpredictable, and constantly requires a judgment nobody wrote down.

And there's a second-order point specific to agtech. The recommendations precision agriculture produces — optimal seeding rate, nitrogen timing, spray decisions — are exactly the kind of constrained optimization problem that automates well. The person interpreting the data is more replaceable than the person deciding, at 5am in the rain, whether the field is fit to travel.

One qualification. This is a statement about the *analysis* role. An agronomist who walks fields, knows a specific farm over years, and is accountable for a recommendation that costs money is doing protected work. One who produces reports from platform data is not. That distinction is the compass for every move from here.

Why production work holds — the six factors



Here's how field farming and crop production answers them. Knowing *why* the field-based work holds at 4.0 while the data layer sits at 7.0 is what tells you which way to move.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

The planning, the monitoring and the record-keeping. Not the doing.

Yield forecasting and field mapping. Input optimization. Compliance and traceability records. Market timing analysis. Equipment telematics and maintenance scheduling. Livestock monitoring and alerting.

What resists: physically operating in a field; handling livestock; fixing equipment where it broke; judging conditions; and everything that happens when the plan meets weather. Those are the things to be moving toward.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Straightforward in production work. Farm labor is in persistent shortage, entry is accessible, and the route in doesn't require a degree. The economics section below covers the harder question, which is what the job pays.

But "easy to enter" hides a fork worth seeing now. The accessibility is real — and it is easiest of all at the two ends that lead nowhere good: small-scale farm ownership, where the median household loses money farming, and the analytical roles that score highest as automation lands on them. The way through is to land on the side that both pays a wage and holds physically — equipment technology, field-based agronomy, the regulated roles — rather than at whichever rung happens to be hiring, or on the romantic assumption that farm ownership is the destination. A seat being open is not the same as a future there.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

Almost entirely, in production roles — and in conditions that vary constantly, which is the harder case for automation.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

Moderately. Food safety, animal welfare and chemical application all carry regulated accountability, and that matters more in livestock and agronomy than in arable production.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

Partially. Pesticide application licensing, veterinary medicine restrictions, food safety certification and organic certification all require qualified humans — a real but uneven protection.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

Constantly. Weather, disease, market shifts, equipment failure. Agriculture is a sequence of unforeseen problems in a fixed calendar. The distance between the field and the screen is the distance you're trying to hold.

The economics, and this is the real conversation

Nothing in this guide matters as much as this section.

USDA's Economic Research Service forecasts the following for 2026:

- Net farm income across the sector: **\$153.4 billion**, down 0.7% nominally and 2.6% after inflation — but still **above the twenty-year average** in real terms
- Net cash farm income: **\$158.5 billion**, up 3.0%
- Direct government farm payments: **\$44.3 billion**, a \$13.8 billion increase on 2025

And then the household figure, which is the one that matters most to where you aim from here:

Median farm income earned by farm households is forecast at $-\$1,161$ for 2026. Negative.
Median off-farm income is forecast at $\$92,815$.

Read that carefully. The median American farm household loses *money farming* and earns nearly \$93,000 doing something else. USDA states plainly that many farm households primarily rely on off-farm income.

The Census of Agriculture makes the same point differently: only 43% of US farms had positive net cash farm income in 2022.

Three things follow, and they should shape your next move more than any score.

Farm ownership at small scale is not a job — it's a business that frequently doesn't pay. That is not a criticism of it; plenty of people choose it knowingly, subsidized by other income. But it's worth knowing before you build a plan around it.

Scale is the determining variable. The 6% of farms with sales over \$1 million produce more than three-quarters of output. Agriculture rewards capital and scale more than almost any sector.

And government payments are a structural feature, at \$44.3 billion in 2026. Farm economics are policy-dependent in a way that few careers are, and that dependency is itself a risk.

Which is why the useful framing is: "farming" and "working in agriculture" are different careers. The second one — equipment technology, agronomy, animal health, food safety, agribusiness — pays a wage and is where most agricultural employment actually is.

What the trend shows

Livestock husbandry: 2.9 → 3.2 → 3.4. Up 0.5 in three years.

Agricultural data and supply planning: 5.8 → 6.5 → 7.0. Up 1.2.

The gap widened from 2.9 points to 3.6. The screen-based end of agriculture is moving more than twice as fast as the field-based end — the same pattern we find in construction, transport and life sciences. Which is why drifting toward the modern-sounding analytical roles is the one direction not to move.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Physical / embodied	Strong in production roles	The one to watch	Agriculture has automated physical work for a century. Autonomous machinery is advancing, and controlled-environment growing removes the unpredictability entirely.
Judgment under novelty	Strong	Durable	Weather, disease and equipment failure are constant and unscripted.
Regulatory / licensing	Partial	Moderate	Chemical application, animal health and food safety require qualified people.
Human trust / accountability	Moderate	Moderate	Matters most in animal welfare and food safety.

The honest read, and it differs from most guides. Agriculture's physical protection is real today and has the longest record of erosion of anything we score. Mechanisation removed three-quarters of this workforce over fifty years. Assuming it has finished would be the least supported assumption in this index.

And there's a specific mechanism to watch, the same one that threatens the skilled trades: **controlled-environment agriculture** — glasshouse, vertical and indoor growing — moves production out of unpredictable fields into structured settings where automation performs well. It is the agricultural equivalent of prefabrication.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Deciding seeding rates and inputs by experience	Platform recommends; the farmer judges whether it fits this field
Scouting fields on foot	Imagery flags the anomaly; someone goes and looks
Keeping compliance records	Automated
Monitoring livestock by observation	Sensors alert; the stockperson assesses
Timing market sales	Analysis assisted
Operating in the field	Largely unchanged
Fixing the machine that broke	Unchanged, and getting more technical

One genuinely growing area. Agricultural equipment has become highly electronic — GPS guidance, variable-rate systems, telematics, and increasingly electrified drivetrains. Equipment technicians who can diagnose both mechanical and electronic faults are in persistent shortage and score 3.8. It is physical, diagnostic, unpredictable work — the protected profile exactly, and one of the most available moves from where you stand.

Human strengths that will matter most

1. **Judging conditions** — whether the field will travel, whether the animal is unwell before

any sensor says so. The least automatable thing here.

2. **Fixing things where they broke** — increasingly a mechanical and electronic skill combined.

3. **Animal handling and stockmanship** — physical, tacit, and learned over years. 4. **Deciding under weather and market uncertainty**, where the data can't tell you. 5. **Accountability for welfare and food safety**, which is regulated and personal.

Notably absent: record-keeping, routine monitoring and input calculation. These were substantial parts of a farm manager's week and are the fastest-depreciating. If your current value is mostly on that second list, the work is to shift it onto the first.

And underneath all five, two traits the score can't grade: **curiosity and adaptability**. The specific tools — the guidance system, the sensor platform, the diagnostic software — will turn over many times across a career. The people who compound are the ones who stay genuinely curious about how a farm actually works and re-tool fastest, rather than defending the one way of working they learned. In a sector that has automated continuously for a century, that flexibility is itself a form of protection — and increasingly it's what employers are hiring for: the trajectory, not the current skill set.

Other things to consider about this job

What's genuinely good about it

Agricultural employment is stable and slightly growing in the wage-and-salary sector — from 1.13 million in 2013 to 1.17 million in 2023, up 4% — even as farm ownership consolidates.

Labor is in persistent shortage, and paid farm labor now accounts for 41% of all farm workers.

And the demographic picture has genuinely improved. Producers under 45 rose to 22% of the total in 2022, up from 20% in 2017, with a documented rise in new and beginning producers and record FFA membership. That reverses a long decline.

What people in agriculture report as rewarding is unusually consistent: working outdoors and with animals; visible, seasonal results; genuine autonomy; the technical variety of modern farming; and a way of life rather than only a job.

And the sector-level income picture is better than the household one — net farm income above its twenty-year average in real terms, with strong receipts.

What's genuinely hard about it

The economics, covered above, and they dominate everything. Median farm household income from farming is negative; 43% of farms have positive net cash income; most farm households rely on off-farm earnings.

Capital requirements are extraordinary. Land, machinery and inputs make entry into ownership nearly impossible without inheritance or substantial backing — which is why the average producer age has been high for decades.

The work is physically demanding and dangerous. Agriculture has among the highest occupational fatality rates of any sector.

Weather and market risk are uninsurable in practice, and a bad year is genuinely a bad year.

And it's policy-dependent — \$44.3 billion in direct government payments in 2026 is a structural feature, and policy changes with governments.

Who this work suits — and who it doesn't

Agriculture tends to suit people who:

- **Genuinely want to work outdoors**, in all conditions, permanently
- **Are practically and mechanically capable** — this is most of modern farm work
- **Can handle uncertainty they don't control** — weather, disease, prices
- **Are comfortable with animals** if livestock is the direction
- **Think in seasons** rather than in weeks

It's harder for people who:

- Need **income predictability**
- Want **defined working hours** — livestock and harvest don't negotiate
- Lack **access to capital or land** and want to farm rather than work in agriculture
- Are drawn to the **idea** of farming without having done it in February

If some of that second list sounds like you, it isn't a reason to abandon the path — it's a reason to steer toward the parts of it that fit and pay, which is what the rest of this does.

What else is moving this market

Consolidation is the dominant structural force and has been for decades. Scale wins.

Controlled-environment agriculture is growing, and it is the most credible long-term threat to the physical protection described here.

Renewable energy is a genuine diversification — 153,101 farms now use renewable systems, up 15%, with three-quarters using solar. Land-based energy income is meaningfully changing farm economics in some regions.

And technology adoption keeps rising — farm internet access went from 75% to 79% between 2017 and 2022 — which grows the agtech sector even as it exposes the analytical roles inside it.

The AI-native advantage — what it actually looks like

The situation: agriculture is the most automated sector in this index by history, and the current wave is landing on its analytical layer rather than its physical one. Being already in it, close to the work, is a real advantage here — you can build this fluency inside a live placement, a first job, or the machinery shed, which is exactly where it counts.

WHAT AN AI-NATIVE AGRICULTURAL PROFESSIONAL LOOKS LIKE

1. Keep the hands and add the diagnostics. Equipment technology is the standout opportunity — modern machinery needs people who can diagnose mechanical, hydraulic and electronic faults together, and they are short everywhere.

2. Own the judgment, not the report. Platforms produce recommendations. The value is in knowing when a recommendation doesn't fit this field, this season, this soil.

3. Verify. A yield model or a spray recommendation that's confidently wrong costs real money in a single application.

4. Take the regulated roles seriously. Chemical application licensing, animal health, food safety and organic certification all require qualified humans and are under-applied to.

5. Understand scale honestly. The economics reward capital. Someone entering agriculture without land should think about which role in the supply chain they want rather than assuming farm ownership.

All five are available to you now — not after some future qualification. Turning them into concrete moves is the rest of this guide.

Where agricultural work can lead later

Farm management, equipment technology and dealership service, agronomy and crop advisory, livestock and animal health, food safety and quality assurance, agricultural supply and input sales, agribusiness and commodity trading, rural land management, policy and extension, and agricultural education.

The pattern in this index holds: the further from the field and the animals, the higher the exposure. Supply planning and data roles score 7.0 against livestock husbandry at 3.4. That's the whole map, and it's the compass for every choice below.

Repositioning from here

The program-selection stage is behind you. The useful questions now are different — and the good news bears repeating: **almost none of your exposure is fixed yet.** Agriculture is a 3.6-point spread from 3.4 to 7.0, and where you land inside it is decided mostly by things still in front of you. Nearly all of them point the same way: toward the hands, and away from the screen.

If you're still in the degree or training

Three levers, in order of leverage.

1. The physical, wage-paying side is the whole game. The exposed rows are precision agriculture technology (6.2) and agricultural data and supply planning (7.0) — screens and data. The protected destinations — livestock husbandry (3.4), equipment technology (3.8), field farming (4.0), farm management (4.6) — are reached by the hands, not the platform. If you haven't committed to a track, this is the single highest-leverage decision left. And work a full season, including winter, before you lock in — a February morning answers the question faster than anything else.

2. Equipment and machinery diagnostics, while it's still free. This is the standout opportunity in the whole field: modern machinery needs people who can diagnose mechanical, hydraulic and electronic faults together, and they are short everywhere. It scores 3.8 — physical, diagnostic, unpredictable, the protected profile exactly. Every hour of real diagnostic depth you can add now is a step from the exposed end toward the protected one, and far cheaper to acquire in training than to bolt on later.

3. The regulated certifications, and the judgment rather than the report. Chemical application licensing, animal health, food safety and organic certification all require qualified humans and are under-applied to. And get fluent with the precision-ag platforms as an operator who *owns the judgment* — knowing when a recommendation doesn't fit this field — rather than as someone who produces the report. **That is the experience that separates the protected field-based end from the exposed analytical floor.**

If you've graduated and you're job-hunting

Here the role behind the title matters more than it looks, because the most modern-sounding early-career titles are also the most exposed.

- **"Precision agriculture" and "ag data / supply planning" roles are the 6.2 and 7.0.** They sound like the future and they're the automating layer at once. Taking one isn't fatal — but take it as a deliberate stepping-stone with an exit in mind, not as the plan.

- **Aim at equipment technology (3.8), field-based agronomy, and livestock and animal health.** Physical, diagnostic, chronically short of people — and under-applied-to, which means less competition, not more.
- **Distinguish *farming* from *working in agriculture* before you accept anything.** If farm ownership is the goal and there's no land or capital, look squarely at the numbers: the median farm household loses money farming and earns \$92,815 off the farm, and only 43% of farms have positive net cash income. Working in agriculture pays a wage; small-scale ownership frequently doesn't.
- **Chase the seat closest to the physical, unpredictable work.** Field-based, hands-on, diagnostic — even at a smaller operation, where responsibility and technical variety tend to arrive sooner than in a data seat at a large agribusiness.

The AI-native move, from where you stand

You can't undo the track you're already on, but you can be the person who directs these tools rather than competes with them. The five capabilities above — keep the hands and add the diagnostics, own the judgment, verify, take the regulated roles seriously, understand scale — are all available right now, inside a current placement or a first job. The most powerful for someone already in it is the first: be the technician who diagnoses mechanical and electronic faults together, or the field agronomist who owns the recommendation rather than producing the report. In agriculture the person deciding at 5am whether the field will travel is more protected than the one interpreting the data — and that's a position you can start building toward this season, not after some future qualification.

The career-hunting checklist

The analysis tells you where the exposure is. This is what to do about it — the part that's actually in your hands. Most of the protection in agriculture is built here, in these choices and habits, rather than handed to you. The first list is for anyone still in the degree or training; the second is for the job hunt itself.

During the degree / training — build what can't be automated

- **Aim at the physical, wage-paying side of the field.** Equipment technology (3.8), field-based production, livestock and animal management (3.4), the regulated roles — over precision-agriculture analytics (6.2) and agricultural data and supply planning (7.0), which are the exposed lanes. The screen-based end scores worst.

- **Work a full season, including winter, before you commit to a track.** Agriculture is romanticized more than almost any career in this index; a February morning answers the question faster than anything else.
- **Get real technical depth in equipment and machinery.** Mechanical, hydraulic and electronic diagnostics together — that's where the shortage is, and it scores 3.8. Physical, diagnostic, unpredictable: the protected profile exactly.
- **Take the regulated certifications seriously.** Chemical application licensing, food safety, animal welfare and organic certification are under-applied to and genuinely protective.
- **Learn the business honestly.** Farm financial management, given a median household income of -\$1,161 from farming, is the most under-taught essential subject in agriculture. If it's an optional module, take it anyway.
- **Get fluent with precision-ag platforms as an operator, not a report-producer.** Be able to direct them, and know when a recommendation doesn't fit this field, this season, this soil. Owning the judgment is protected work; producing the report is not.
- **Feed curiosity and adaptability on purpose.** The guidance systems and sensor platforms will turn over repeatedly; the person who re-tools fastest, and stays genuinely interested in how a farm works, compounds.
- **Keep the frame wide.** Properly vet equipment technology, field-based agronomy, veterinary and animal science, and food safety before ruling any of them out. The modern-sounding lane is not the safe one here.

During the job hunt — aim, don't just apply

- **Target the protected lanes on purpose.** Equipment technology (3.8), field-based agronomy, livestock and animal health, the regulated roles — not "precision agriculture analyst, open to anything," which points straight at the 6.2-7.0 end.
- **Read the role, not just the title.** A "precision agriculture" or "ag data" seat is the analytical layer (6.2 and 7.0). If you take one, take it as a deliberate stepping-stone with an exit in mind, not as the plan.
- **Don't grab a role just because it's on offer in an exposed lane.** A role being available doesn't make it safe. An employer still hiring analysts to produce recommendations from platform data may simply be behind the curve — and that's exactly the rung that gets cut first once they catch up. Availability can be a lagging indicator, not a green light.
- **Distinguish farming from working in agriculture before you accept anything.** If the goal is farm ownership, be honest about land and capital early: the median farm household loses money farming and earns \$92,815 off the farm, and only 43% of farms have positive

net cash income. Working in agriculture pays a wage; small-scale ownership frequently doesn't.

- **Prefer the seat closest to the physical, unpredictable work.** Field-based, hands-on, diagnostic — the field that doesn't match the map, the animal that's unwell, the machine that broke — over the seat that produces recommendations from a screen.
- **Ask the employer where AI is heading in the role.** Whether they're building the job around autonomous machinery and platforms or waiting to be forced to tell you if the seat grows or is quietly being hollowed out.
- **Weigh the trajectory over the starting pay.** A physical, diagnostic role that compounds beats a higher-paid analytical seat that's being automated underneath you.

Go deeper — further reading

- **Farm Sector Income Forecast** — USDA Economic Research Service. The primary source for every income figure in the economics section, including the median farm household figures. Free, updated regularly, and the most important reading on this page.
- **Census of Agriculture** — USDA National Agricultural Statistics Service. The structural picture: farm numbers, scale concentration, producer demographics, technology adoption. The 6%-of-farms-produce-75%-of-output finding comes from here.
- **Farm Labor** — USDA ERS. The seventy-five-year employment history described above, and current workforce data.

The counterargument — read this one:

The strongest case against our field-farming score is agriculture's own history.

This sector shed three-quarters of its workforce while tripling output. Assuming the physical work is now safe because current robots handle it poorly ignores that agriculture has automated physical work continuously for a century — and that autonomous tractors, robotic milking and automated harvesting are all in commercial deployment today, not in laboratories.

On this reading our 4.0 reflects a snapshot and misses the trajectory, and the sector that automated most thoroughly is unlikely to have stopped.

Where we land: we think this is the strongest historical counterargument to any score in the index, and it's why we rate physical protection as the least durable factor everywhere.

What holds our score is that the remaining work is disproportionately the *unstructured* part — the field that doesn't match the map, the animal that's unwell, the machine that broke somewhere inconvenient. Automation removed the repetitive physical work first and has been slower with the rest.

But the honest position is that agriculture is where our physical-protection assumption has been tested longest and held least well. It's worth weighting that history heavily.

Bottom line

Agriculture is the field where what you do *from here* matters more than the subject itself — which is exactly why being already in it is not a problem: the decisive moves were always going to happen from where you stand.

Three-quarters of the workforce disappeared while output tripled. The work survived; the jobs consolidated; the survivors got bigger rather than safer. That happened over fifty years, through machinery rather than software, and nobody outside the sector noticed.

The current wave is landing on the analytical layer, not the physical one. Precision agriculture scores 6.2 and data planning 7.0, against livestock husbandry at 3.4 and field production at 4.0. The tech-forward version of this career is the more exposed one — so the move from here is toward the hands, not the screen.

But the real question was never automation. USDA forecasts the median farm household will lose money farming in 2026 and earn \$92,815 off the farm. Only 43% of farms have positive net cash income. Six per cent of farms produce more than three-quarters of output.

So the useful advice isn't a verdict on the path — it's a direction. Distinguish *farming* from *working in agriculture*, because the second pays a wage and is where most agricultural employment is. Steer hard toward equipment technology, which is physical, diagnostic, chronically short of people and scores 3.8. Take the regulated roles seriously. And if the goal is to farm rather than to work in agriculture, be honest about capital and land now rather than later.

And one thing worth saying to someone genuinely drawn to this. *Working outdoors, with animals or land, in a sector where results are visible and seasonal, is something very few careers offer at all. The economics are hard and the history is instructive — and neither of those is the same as saying you're in the wrong place.*

Questions worth sitting with

Whether you're the one in it or a parent thinking it through, these are the conversations that move the needle now. Read "you" as whoever the decision is about.

Where you actually stand

1. Which track are you on — or aiming at — and does it point at the *hands* (field production, livestock, equipment) or at the *screen* (precision analytics, data and supply planning)? 2. Looking at the score table, which destination are you currently pointed at, and is it the one you'd choose on purpose? 3. **Do you want to farm, or to work in agriculture?** Different careers, different economics — and if it's farming without land or capital, have you looked at the median household losing money farming and earning \$92,815 off the farm?

The move from here

4. What's the single most protected role you could realistically aim for from where you are now — equipment technology, field-based agronomy, animal health — and what would make you a credible candidate for it? 5. If you're still studying: how much real equipment and machinery diagnostics are you getting, and is there more you could add before it's no longer free? 6. If you're job-hunting: are you aiming at a protected, physical lane on purpose, or applying to whatever's open? Name the three roles you're actually chasing.

The AI question, practically

7. Of the five capabilities — keep the hands and add the diagnostics, own the judgment, verify, take the regulated roles seriously, understand scale — which one could you visibly demonstrate within the next few months? 8. Can you point to a moment where you knew a platform's recommendation didn't fit the field, and were right? If not, where's the easiest place to start building that judgment?

Testing it against reality

9. **The most valuable single action:** work a full season on a commercial operation, including the parts nobody photographs — and find someone working in agriculture (a technician, an agronomist, a food safety manager) to compare what they earn and what their week looks like. 10. If the first offer that comes is a "precision agriculture" or "ag data" role, what's your plan — take it as a deliberate stepping-stone with an exit in mind, or hold out for something closer to the field? Decide before the offer, not after.

This is analysis and informed judgment about a fast-moving situation. For agriculture specifically, we've flagged that this sector has the longest history of automation of anything we measure, that our physical-protection assumption has been tested here longer than anywhere and held least well, and that the decisive issues are economic rather than technological.

Scores for 2023 and 2025 are retrospective reconstructions using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Agriculture

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned}\text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment})\end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-eight careers, which is what makes the scores comparable.

Task automatability (A) — *what share of the working week could a capable current system already do to an acceptable standard?*

RATING	ANCHOR
0–2	Almost nothing. The work is physical, relational or wholly novel.
3–4	Administrative surround only — notes, scheduling, correspondence.
5–6	A meaningful minority of the week, mostly documentation and lookup.
7–8	A large share, including tasks central to the junior version of the role.
9–10	Most of the described duties, at or near professional standard.

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
0–2	Protected by statute or physical necessity. Training hours cannot be automated.
3–4	Stable. Some junior tasks absorbed; the apprenticeship survives.
5–6	Visible narrowing. Employers prefer experience; junior intake reduced.
7–8	Substantial. The work juniors learned on has largely gone.
9–10	The training layer and the automatable layer were the same layer.

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
0–2	Fully screen-based. Location-independent.
3–4	Occasional presence; the core work is not physical.
5–6	Regular physical presence, in largely controlled conditions.
7–8	Substantially physical, with some variability of setting.
9–10	Irreducibly physical, in conditions that differ every time.

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No license exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

LIVESTOCK HUSBANDRY & ANIMAL MANAGEMENT — 3.4 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.4	5.2	5.7	1.99
Entry-path erosion	15%	1.8	2.0	2.2	0.33
Physical / embodied	15%	9.5	9.5	9.5	0.07
Human trust / accountability	15%	7.0	7.0	7.0	0.45
Regulatory / licensing	10%	6.0	6.0	6.0	0.4
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	3.4 → 3.4

AGRICULTURAL EQUIPMENT TECHNOLOGY & REPAIR — 3.8 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.5	5.3	6.0	2.1
Entry-path erosion	15%	2.0	2.2	2.5	0.38
Physical / embodied	15%	9.3	9.3	9.3	0.1
Human trust / accountability	15%	6.5	6.5	6.5	0.53
Regulatory / licensing	10%	4.5	4.5	4.5	0.55
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	3.8 → 3.8

FIELD FARMING & CROP PRODUCTION — 4.0 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.0	5.8	6.5	2.27
Entry-path erosion	15%	2.0	2.2	2.5	0.38
Physical / embodied	15%	9.3	9.3	9.3	0.1
Human trust / accountability	15%	6.0	6.0	6.0	0.6
Regulatory / licensing	10%	5.0	5.0	5.0	0.5
Judgment under novelty	10%	8.5	8.5	8.5	0.15
				Total	4.0 → 4.0

FARM MANAGEMENT — 4.6 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.5	6.5	7.2	2.52
Entry-path erosion	15%	2.5	2.8	3.2	0.48
Physical / embodied	15%	7.5	7.5	7.5	0.38
Human trust / accountability	15%	6.5	6.5	6.5	0.53
Regulatory / licensing	10%	5.0	5.0	5.0	0.5
Judgment under novelty	10%	8.0	8.0	8.0	0.2
				Total	4.6 → 4.6

AGRONOMY & CROP ADVISORY — 5.4 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	6.1	7.3	8.3	2.91
Entry-path erosion	15%	3.2	3.6	4.0	0.6
Physical / embodied	15%	5.5	5.5	5.5	0.67
Human trust / accountability	15%	6.5	6.5	6.5	0.53
Regulatory / licensing	10%	6.0	6.0	6.0	0.4
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	5.41 → 5.4

PRECISION AGRICULTURE TECHNOLOGY — 6.2 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.4	6.2	7.4	2.59
Entry-path erosion	15%	4.0	4.5	5.0	0.75
Physical / embodied	15%	2.5	2.5	2.5	1.12
Human trust / accountability	15%	5.5	5.5	5.5	0.67
Regulatory / licensing	10%	3.5	3.5	3.5	0.65
Judgment under novelty	10%	6.0	6.0	6.0	0.4
				Total	6.19 → 6.2

AGRICULTURAL DATA & SUPPLY PLANNING — 7.0 (HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.9	6.6	7.8	2.73
Entry-path erosion	15%	4.5	5.2	5.8	0.87
Physical / embodied	15%	1.0	1.0	1.0	1.35
Human trust / accountability	15%	5.0	5.0	5.0	0.75
Regulatory / licensing	10%	2.0	2.0	2.0	0.8
Judgment under novelty	10%	5.0	5.0	5.0	0.5
				Total	7.0 → 7.0

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Livestock husbandry & animal management

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	4.4	1.8	1.07	2.9
2025	5.2	2.0	1.07	3.2
Fall 2026	5.7	2.2	1.07	3.4

Movement decomposition: automatability contributed **+0.45** and entry-path erosion **+0.06**, for a total of **+0.5** points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labor market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.